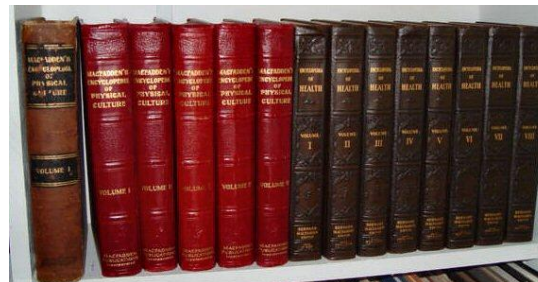


Programmable Machines



Programmable Machines

- Computers are **programmable**
 - Unlike other machines, it's cheap and easy to tell a computer how to solve a new problem
 - Even a problem that no one had thought of when they made the computer!
 - This is what makes computing such a powerful tool

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 - This is what makes computing such a powerful tool
- Computer Science is the **science** and **practice** of **solving problems** using **computing**
 - The Science:
 - What is computing? How does it work? What can and can't it do?
 - The Practice:
 - How to you solve real-world problems with computing?

What Problems?

- What kind of problems do you want to solve?
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- What kind of problems do you want to solve?
 - Can computing help with that?
- Computing is processing data
 - Data (numbers, text, images, videos)
 - Input (user interaction, data sources, sensors)
 - Output (screens, audio, electronics including other computers)
- Examples: alarm, text messages, drone flight

What Problems?

- Imagine you have a touchscreen that you can put on any object in your house, turning it into a computer
 - It can have sensors, connect to the internet, control other devices
- Example: digital thermostat
- What kinds of problems could you solve?
 - What data is involved and how does the computer use it?

What Problems?

- Imagine you have Augmented Reality (AR) glasses that can display objects as if they were in the real world
 - You can connect to wearable sensors, the internet, control other devices
- Example: friend locator
- What kinds of problems could you solve?
 - What data is involved and how does the computer use it?

What Problems?

- Over 90% of Americans carry a smartphone
 - Everybody has a computer with them!
- We have smart homes, cars, offices, schools, hospitals...

What kinds of problems do you want to solve?

Artificial Intelligence

- The big topic right now
 - What is “Generative AI”? How does it work?
 - What problems can it solve?
 - What are its limitations? Costs? Dangers?
 - As a student, how should you approach AI?
 - What should your instructors be doing with it?
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- A Computer Scientist is an expert on computing
 - You have to always be learning
 - Staying ahead of it is a big challenge